Field Effect Transistors

- Introduction.
- The MOSFET.
- Current in MOSFETs and Operation Modes.
- DC Analysis (Biasing).

Introduction

From diodes to transistors

Transistors have an extra connection (3-terminal devices)

Basic applications of transistors: 1) switching 2) signal manipulation

Transistor Basic Types: FET and BJT

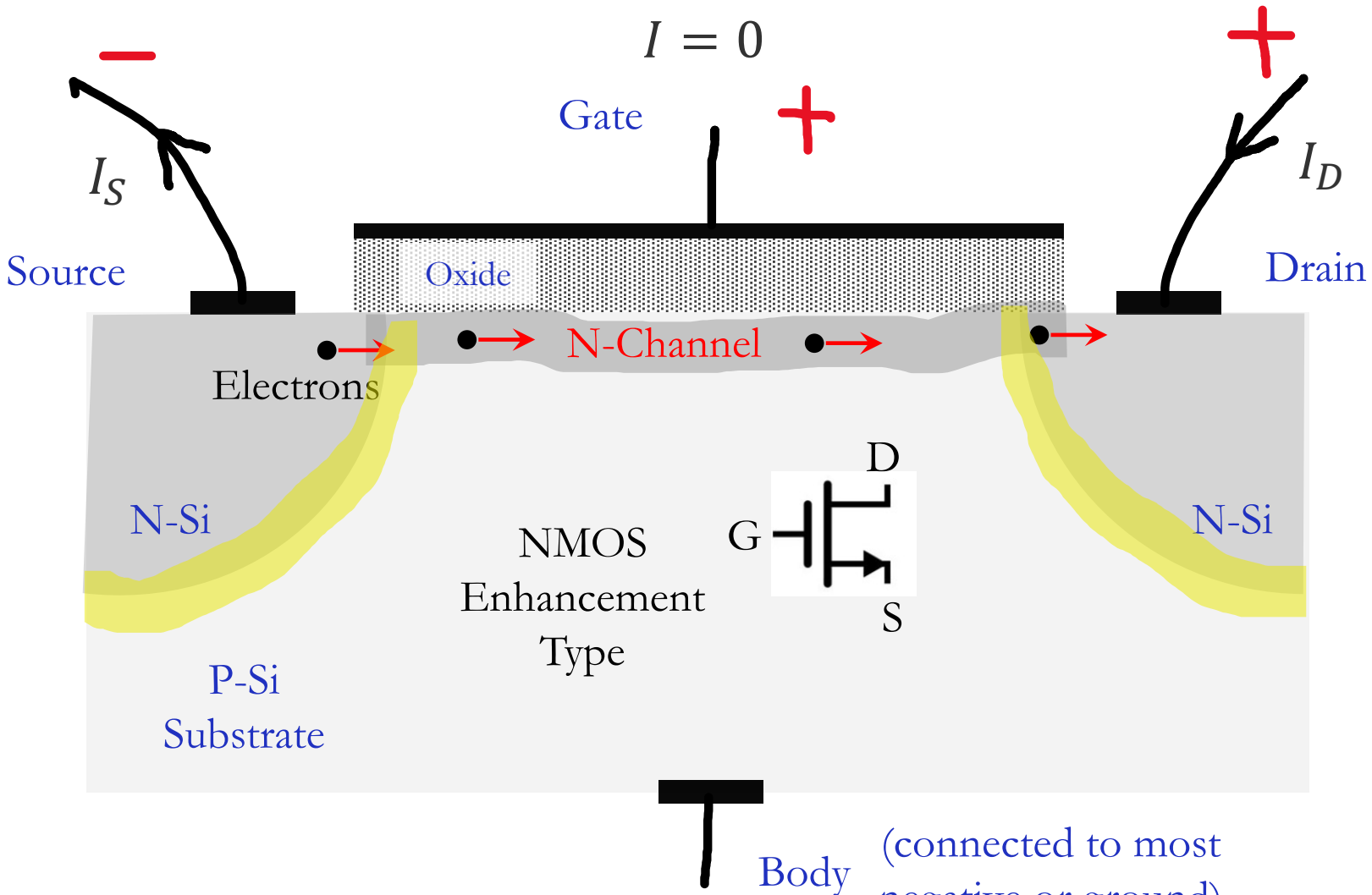
The FET family: JFET, MESFET, MOSFET, CMOS

MOSFET come with two flavors: NMOS and PMOS

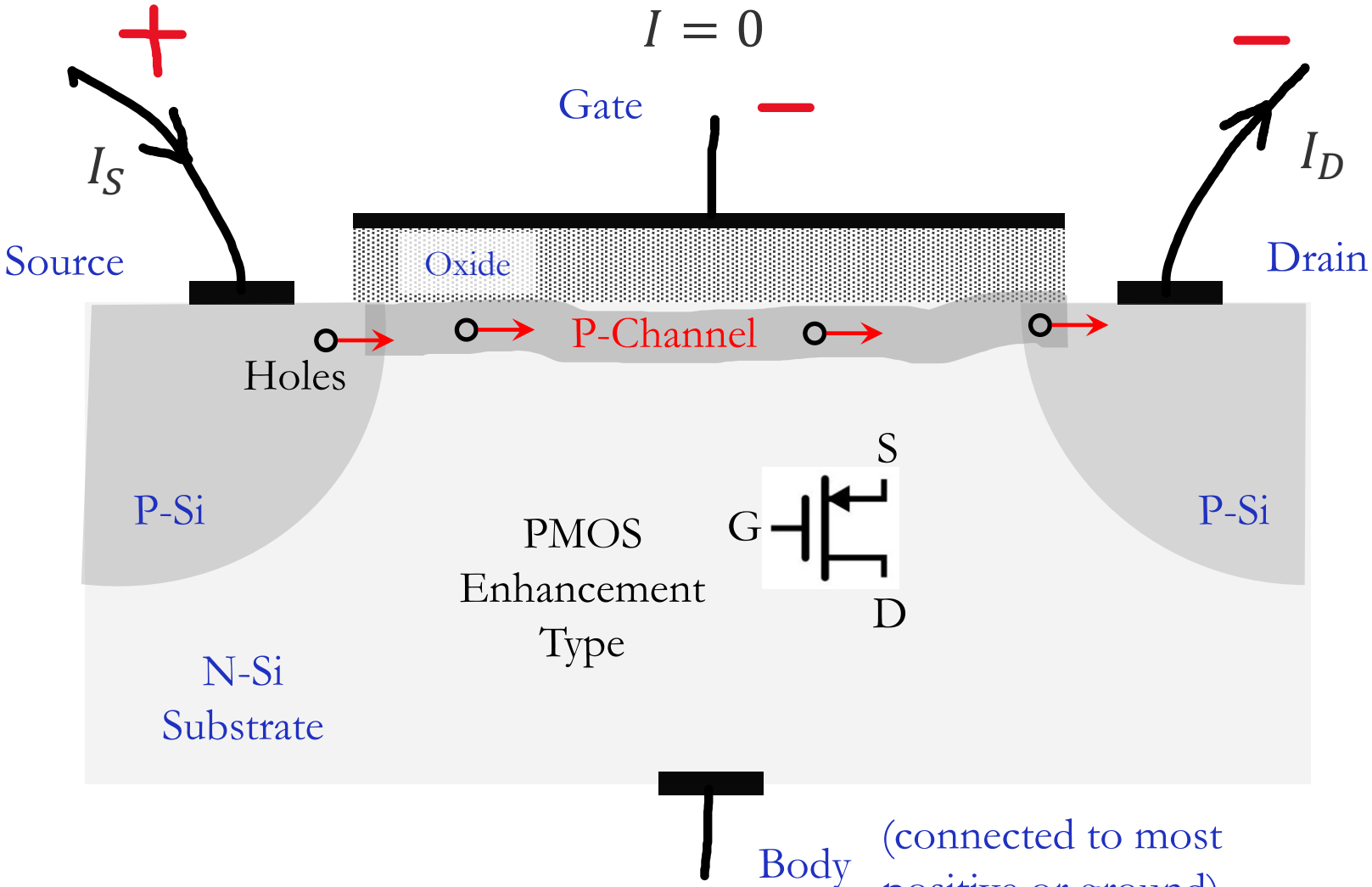
CMOS is the combination of NMOS and PMOS

MOSFET= Metal-Oxide-Semiconductor-Field-Effect-Transistor

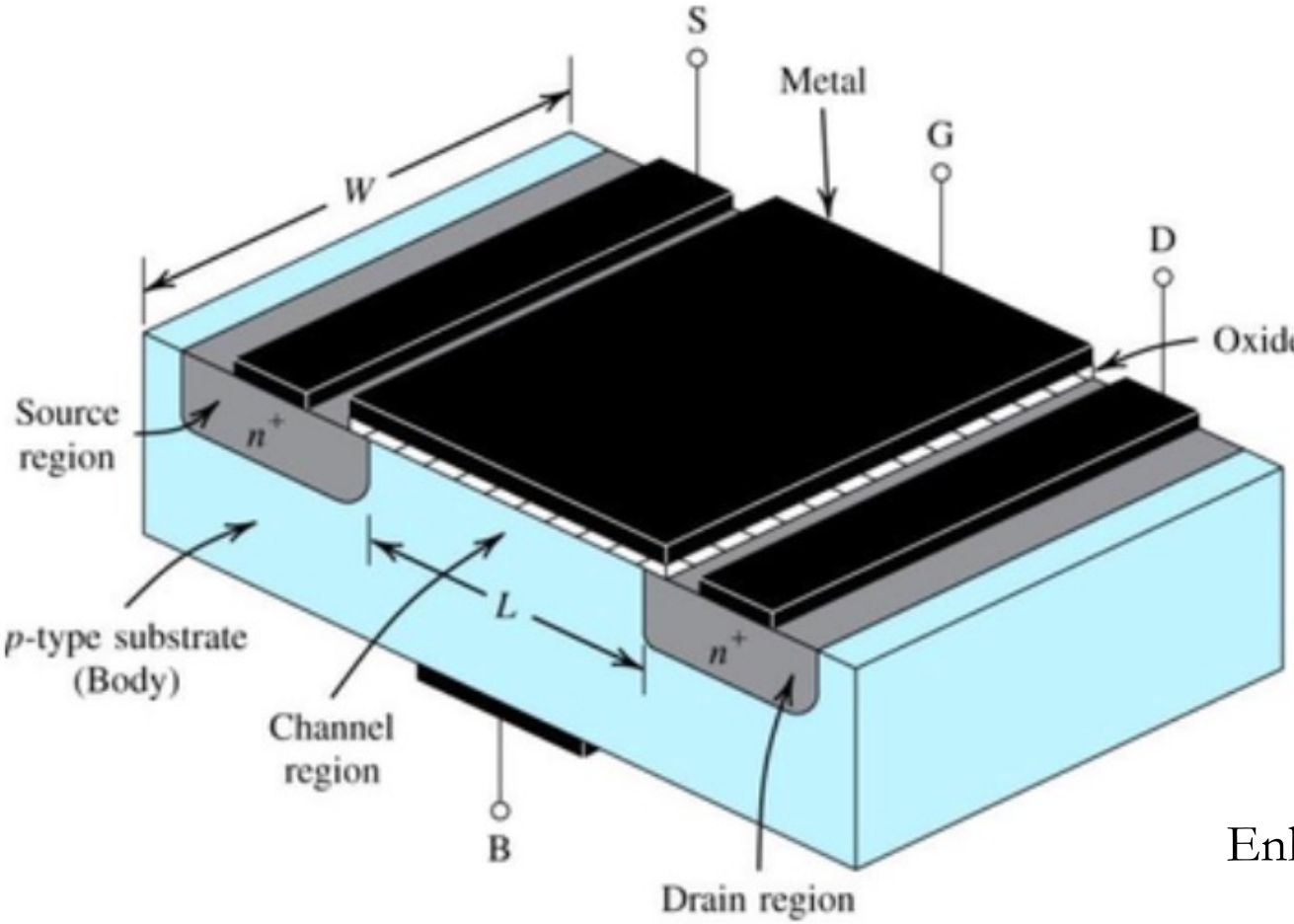
The MOSFET Structure



The MOSFET Structure



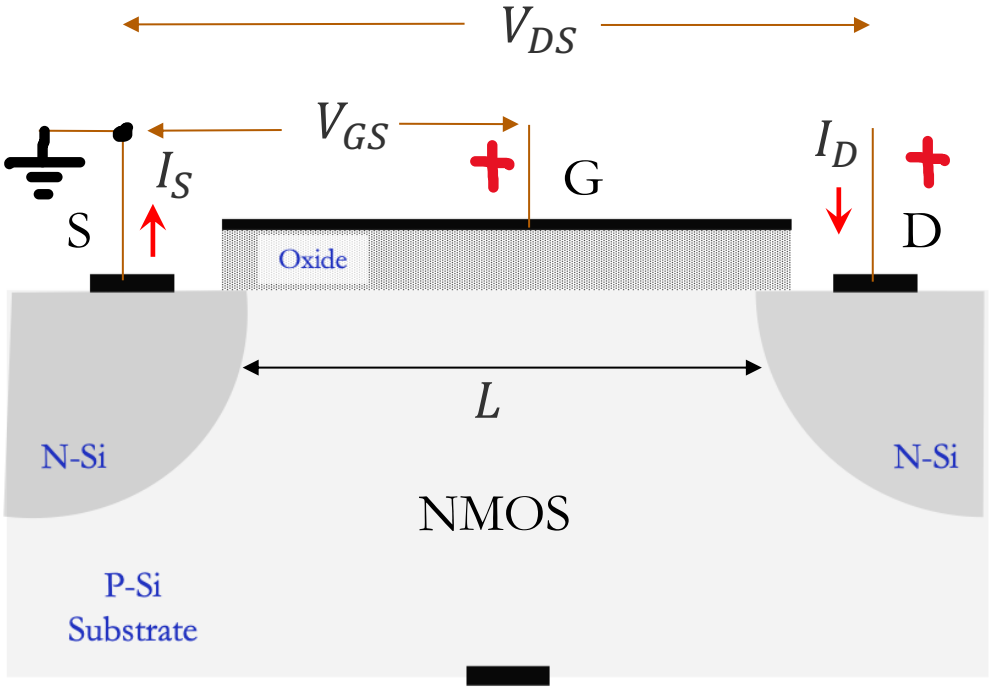
The MOSFET Structure



$$C_{ox} = \frac{\varepsilon WL}{t_{ox}}$$

NMOS Enhancement Type

Current in a MOSFET



To have any current, a channel need to be created.

$$V_{GS} \geq V_t$$

V_t is the threshold voltage. The voltage needed to start the channel. $V_{GS} < V_t$, nothing will happen $I_D = I_S = 0$. This is the Cut-off Mode.

The value of V_t (1 ~ 2 V) depends on:

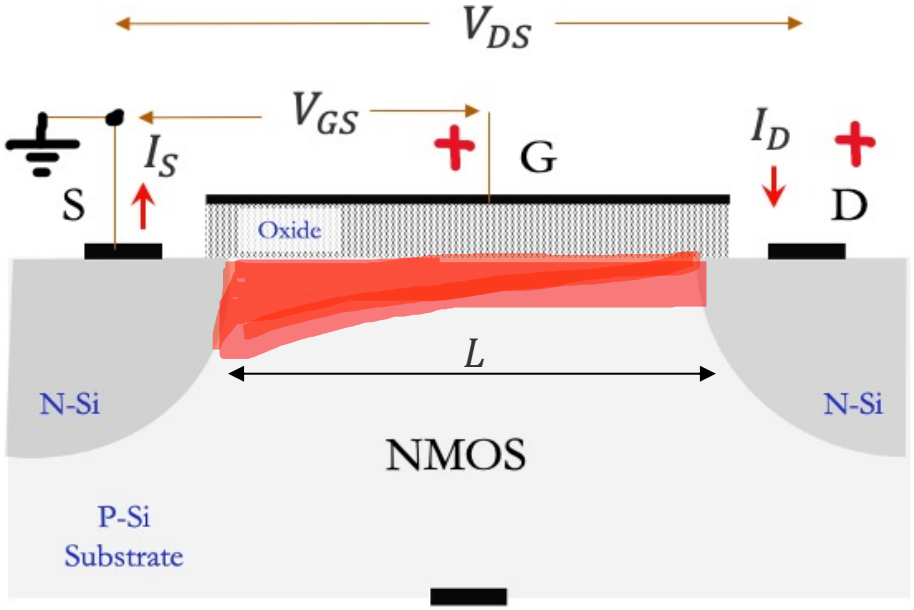
- Substrate Doping.
- Gate Oxide.
- Material.
- Body voltage.

With $V_{GS} \geq V_t$, the current depends on the value of V_{DS}.

Two regions can be identified:

1) $V_{DS} \leq V_{GS} - V_t$ (Triode)
2) $V_{DS} \geq V_{GS} - V_t$ (Saturation)

Current in a MOSFET



Electron mobility — Oxide capacitance

NMOS $$k_n = \frac{\mu_n C_{ox}}{2}\frac{W}{L}$$ → Aspect ratio (size)

PMOS $$k_p = \frac{\mu_p C_{ox}}{2}\frac{W}{L}$$

1) Cutoff Mode

$$V_{GS} < V_t$$

$$I_D = 0$$

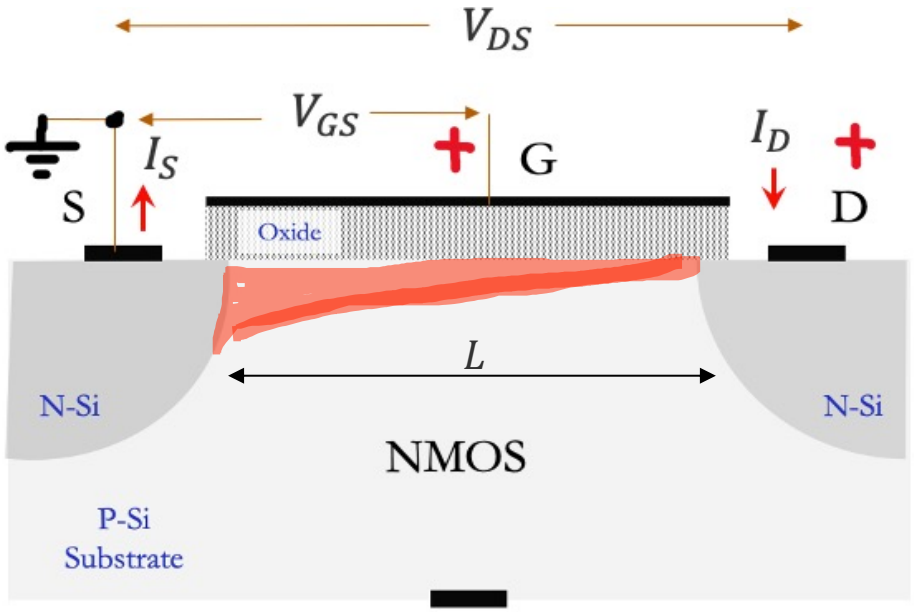
2) Triode Mode

$$V_{GS} \geq V_t \text{ and } V_{DS} \leq V_{GS} - V_t$$

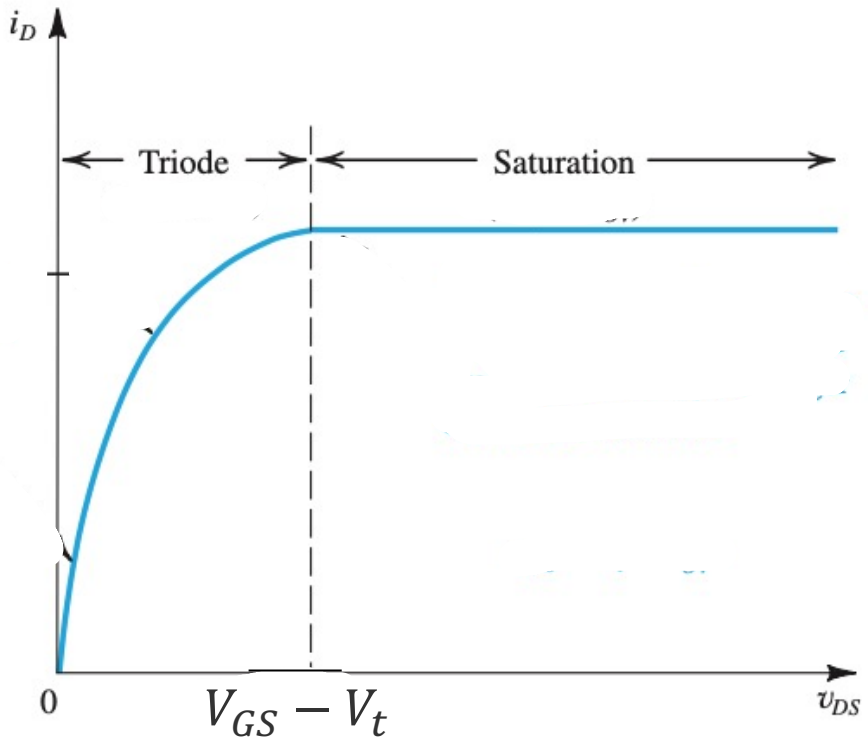
$$I_D = k\left[2(V_{GS} - V_t)V_{DS} - {V_{DS}}^2\right]$$

$$k = \frac{\mu\, C_{ox}}{2}\frac{W}{L}$$

Current in a MOSFET



MOSFET IV Curve



Pinch-off

3) Saturation Mode

$$V_{GS} \geq V_t \text{ and } V_{DS} \geq V_{GS} - V_t$$

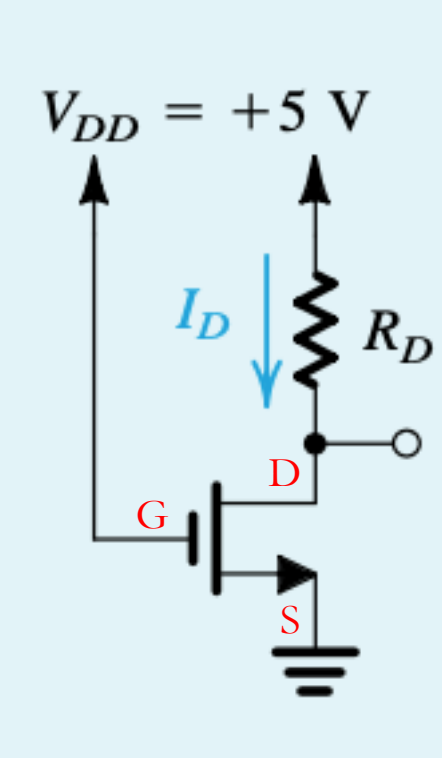
$$I_D = k(V_{GS} - V_t)^2$$

Summary

Mode	NMOS (V_t +ve)	PMOS (V_t -ve)
Cutoff	$V_{GS} < V_t \quad I_D = 0$	$V_{GS} > V_t \quad I_D = 0$
Triode	$V_{GS} \geq V_t$ and $V_{DS} \leq V_{GS} - V_t$ $I_D = k\left[2(V_{GS} - V_t)V_{DS} - {V_{DS}}^2\right]$	$V_{GS} \leq V_t$ and $V_{DS} \geq V_{GS} - V_t$ $I_D = k\left[2(V_{GS} - V_t)V_{DS} - {V_{DS}}^2\right]$
Saturation	$V_{GS} \geq V_t$ and $V_{DS} \geq V_{GS} - V_t$ $I_D = k(V_{GS} - V_t)^2$	$V_{GS} \leq V_t$ and $V_{DS} \leq V_{GS} - V_t$ $I_D = k(V_{GS} - V_t)^2$

Examples

1) Check the transistor operation mode in the circuit below $R_D = 12\ k\Omega, I_D = 0.4\ mA, V_t = 1\ V$.



$$V_S = 0$$

$$V_G = 5\ V$$

$$V_{GS} = 5 - 0 = 5\ V$$

$$V_{GS} > V_t$$

Not Cut-off

$$V_D = 5 - (12k)(0.4\ m) = 0.2\ V$$

$$V_{DS} = 0.2 - 0 = 0.2\ V$$

Check

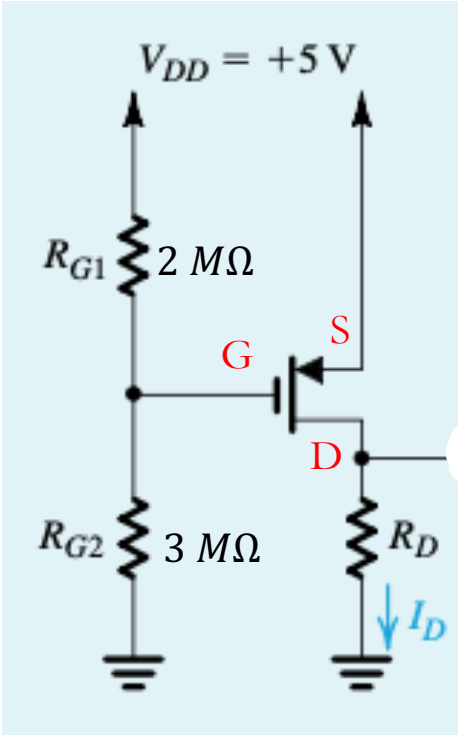
$$V_{DS} \quad ? \quad V_{GS} - V_t$$

$$0.2 \quad < \quad 5 - 1$$

Triode Mode

Examples

2) Check the transistor operation mode in the circuit below $R_D = 6\ k\Omega, I_D = 0.5\ mA, |V_t| = 1\ V$.



$$V_S = 5\ V \qquad V_G = 5\frac{3}{5} = 3\ V$$

$$V_{GS} = 3 - 5 = -2\ V \qquad V_{GS} < V_t$$

Not Cut-off

$$V_D = (6k)(0.5\ m) = 3\ V$$

$$V_{DS} = 3 - 5 = -2\ V$$

Check $\quad V_{DS} \quad ? \quad V_{GS} - V_t$

$$-2 \quad < \quad -2 - (-1)$$

Saturation Mode

$$V_D - V_S \quad ? \quad V_G - V_S - V_t$$

$$3 \quad < \quad 3 - (-1)$$

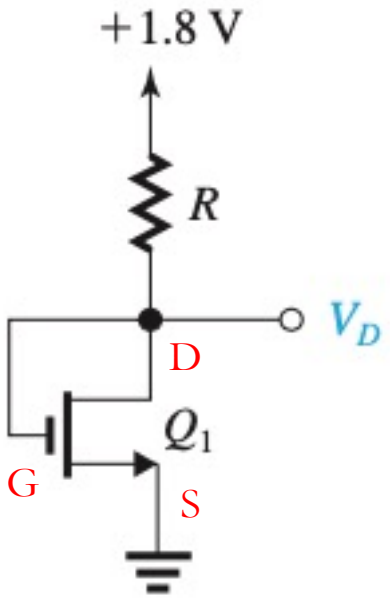
What should R_D be for edge of saturation?

$$V_D = V_G - V_t = 4\ V$$

$$R_D = \frac{4}{0.5m} = 8\ k\Omega$$

Examples

3) Analyze the transistor circuit below. Take $R_D = 13.9\ k\Omega, V_t = 0.5\ V$.



$\mu_n C_{ox} = 0.4\ \text{mA/V}^2$

$\frac{W}{L} = \frac{0.72\ \mu m}{0.18\ \mu m}$

$k_n = 0.8\ \text{mA/V}^2$

Check mode of operation

$V_S = 0 \qquad V_G = V_D$

$V_{DS} \quad ? \quad V_{GS} - V_t$

$V_D \quad > \quad V_G - V_t$

Saturation Mode

$I_D = k(V_{GS} - V_t)^2$

$I_D = 0.8\ m\ (V_G - 0.5)^2 = \frac{1.8 - V_G}{13.9\ k}$

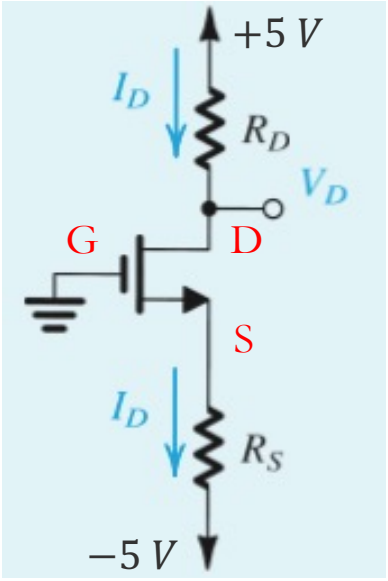
Solve for V_G

$V_G = 0.8\ V = V_D$

$I_D = 0.072\ mA$

Examples

4) Design the transistor circuit below for $I_D = 0.4\,mA, V_D = +1\,V$. Take $V_t = 2\,V$.



Check mode of operation

$$V_G = 0$$

$$V_{DS} \quad ? \quad V_{GS} - V_t$$

$$V_D \quad ? \quad V_G - V_t$$

$$+1 \quad > 0 - 2$$

Saturation Mode

$$I_D = k(V_{GS} - V_t)^2$$

$$0.4\,m = 0.4\,m\,(V_{GS} - 2)^2$$

Solve for V_{GS}

$V_{GS} = +1\,V$ Cutoff ✗

$V_{GS} = +3\,V$ ✓

$V_S = -3\,V$

$\mu_n C_{ox} = 20\ \mu\text{A/V}^2$

$\frac{W}{L} = \frac{400\,\mu m}{1\,\mu m}$

$k_n = 0.4\ \text{mA/V}^2$

$$R_S = \frac{-3 - (-5)}{0.4\,m} = 5\Omega k$$

$$R_D = \frac{5 - 1}{0.4\,m} = 10\Omega k$$

Q & A

Nano Photonics and Plasmonics Research

Send me an email if
you have any
questions.